

STANDARD OPERATING PROCEDURE FOR SPOT WELDING

- Must use safety helmet, safety gloves, safety goggles, and safety shoes
- Adjust the air pressure to suitable value using the pressure gauge for movement of electrodes.
- Set the rotary switch/link switch to low position.
- Set the weld ON /OFF switch to 'weld off' position.
- Turn ON the main electric supply, voltage is set according to the thickness of work piece.
- Neon lamp on the control timer should glow.
- Press the foot switch. The electrodes should close on to each other. Electrodes should be aligned by operator foot switch ON/OFF.
- Two pieces of similar gauge material within the welding range of the machine should be selected.
- 'Squeeze' time on the control timer should be set for approximately 25 cycles.
- Weld time should be set according to the thickness.
- Hold time should be set at least equal to weld time.
- Set weld ON/OFF switch to ON position.
- Ensure there is a flow of water from the cooling tower to inside the electrodes, this helps the electrodes to cool.
- Check the inflow and outflow of water pipes.
- Heat control potentiometer on the control timer should be set at about 50%.
- Press foot switch again. The electrodes will come together and the welding sequence will commence. The current will pass during the weld time fusing the metal pieces together. The electrodes will separate at the end of hold time. Release the foot switch.
- A strong weld would normally be yielded provided the electrode tip area confirms to the recommended value and welding current is adequate in magnitude.
- Check the strength of the weld. If weld is poor, rotate the heat control knob on the control timer clock wise to obtain higher welding current and repeat the operation.
- If the welding is continuous to be of poor quality change the position of the rotary switch/ link switch to the next high position and repeat the operation until you obtain a satisfactory weld.
- If the welding current is increased to too high value, sparking would result. In such case reduce the weld current slightly and increase the pressure.
- Apply lubricants such as Grease on the pathway for forward and backward motions of the spot welding machine.



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